TS127QFM-LL0-DKP0 Product Specification Rev.0

BUYER	Lenovo
SUPPLIER	ORDOS YUANSHENG Optoelectronics Technology CO., LTD
FG-Code	TS127QFM-LL0

ITEM	BUYER SIGNATURE	DATE

ITEM	SUPPLIER SIGNATURE	DATE
Prepared		
Reviewed		
Approved		

ORDOS YUANSHENG Optoelectronics Technology CO., LTD

REVISION HISTORY

REV.	ECN No.	DESCRIPTION OF CHANGES	DATE	PREPARED
P0		Initial Release	2022-01-06	Zhao Shuai

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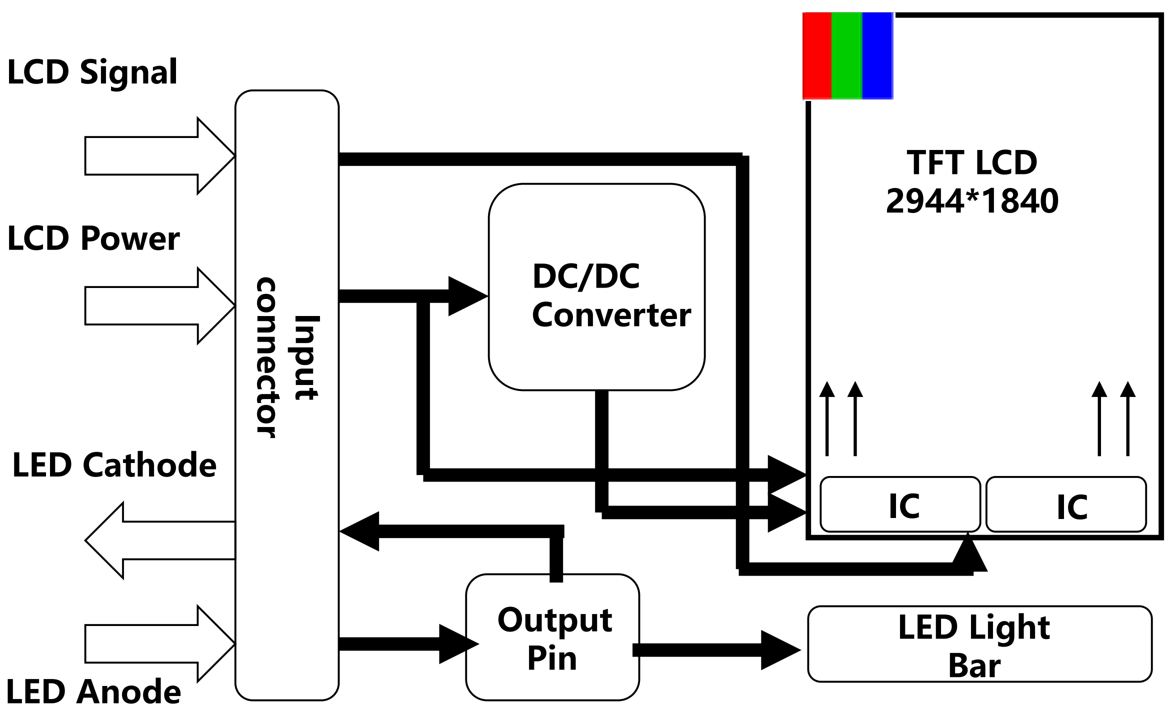
1.0 GENERAL DESCRIPTION

1.1 Introduction

TS127QFM-LL0-DKP0 is a color active matrix LTPS LCD Panel using Low Temperature Poly-silicon TFT ' s (Thin Film Transistors) as an active switching devices. The LTPS-LCD has a 12.7 inch diagonally measured active area with WQXGA resolutions (2944horizontal by 1840vertical pixel arrays). Each pixel is divided into RED, GREEN, BLUE dots which are arranged in vertical stripe and this module can display 16.7M colors.



1.2 Application

- Tablet PC

1.3 General Specification

The followings are general specifications at the TE115C9M-LL0

<Table 1. LCD Module Specifications>

Parameter	Specification	Unit	Remarks
Active Area	273.61536 (H) x 171.0096 (V)	mm	
Number Of Pixels	2944(H) ×1840(V)	pixels	
Pixel Pitch	0.03098(H)×RGB×0.09294(V)	mm	
Pixel Arrangement	Pixels RGB stripe arrangement		
Display Mode	ADS		
Display Colors	16.7M	colors	
Surface Treatment	HC		
Contrast Ratio	1500:1(typ.)1000:(min)		
Viewing Angle(CR>10)	Min. 80°, Typ. 85°	deg.	
Response Time	Max 25	ms	
Color Gamut	93%		
Brightness	Typ.400 Min.320	cd/m2	@Center
Brightness Uniformity	Min.75%		@13P
Power Consumption	LCD: 600mW(Max.)(White Pattern BLU: 4385mW(Max.)	watt	21mA, 2.9Vmax, 72EA W/O LED Driver
Outline Dimension	289.52*186.91*0.5@TLCM 278.02*179.02*1.8@LCM	mm	
Weight	240g (max)	gram	

<Table 2. Touch Panel Specifications>

<table>
<tr><th colspan="2">Parameter</th><th colspan="4">Specification</th><th>Remarks</th></tr>
<tr><td colspan="2">TP Structure</td><td colspan="4">TDDI</td><td></td></tr>
<tr><td colspan="2">Sensing Method</td><td colspan="4">Self Capacitance</td><td></td></tr>
<tr><td colspan="2">Number Of Touch</td><td colspan="4">60*40</td><td></td></tr>
<tr><td rowspan="8">Performance</td><td>Sensitivity(mm)</td><td colspan="4">Φ4</td><td></td></tr>
<tr><td>Report Rate</td><td colspan="4">120Hz</td><td></td></tr>
<tr><td>Finger Separation</td><td colspan="4">≦13mm</td><td></td></tr>
<tr><td>Response Time</td><td colspan="4">≦30ms</td><td>Power up to Active model</td></tr>
<tr><td>Accuracy(mm)</td><td rowspan="4">Center</td><td><=1.0</td><td rowspan="4">Edge</td><td><=1.5</td><td rowspan="4">@Φ6mm</td></tr>
<tr><td>Precision(mm)</td><td><=1.0</td><td><=1.5</td></tr>
<tr><td>Jitter (mm)</td><td><=0.5</td><td><=0.5</td></tr>
<tr><td>Linearity(mm)</td><td><=1.0</td><td><=1.5</td></tr>
<tr><td colspan="2">SNR(dB)</td><td colspan="4">30:1</td><td>@Φ6mm</td></tr>
<tr><td colspan="2">Pen Type & Pen Size</td><td colspan="4">-</td><td></td></tr>
<tr><td colspan="2">Glove Touch</td><td colspan="4">-</td><td></td></tr>
<tr><td colspan="2">Hover Reading Height (center)</td><td colspan="4">Equal or better than WGP level1</td><td></td></tr>
<tr><td rowspan="2">Anti Water</td><td>Spray</td><td colspan="4" rowspan="2">TP works well under below condition,
1,Water-drop(Size φ1mm) on the top of TP.
2,Spray on the top of TP.</td><td></td></tr>
<tr><td>Water Drop</td><td></td></tr>
<tr><td colspan="2">Power Consumption(Max.)
Active/Idle/Sleep</td><td colspan="4">140mW/-mW/-mW</td><td></td></tr>
</table>

2.0 ABSOLUTE MAXIMUM RATINGS

The followings are maximum values which, if exceed, may cause faulty operation or damage to the unit.

< Table 3. Absolute Maximum Ratings>

Parameter		Symbol	Min.	Typ .	Max.	Unit	Remarks
Power Supply	LCD Module	VSP	0		6.5	V	Ta = 25 ℃ Note 1&2
		VSN	-6.5		0	V	
		VDDD	0		-	V	
		IOVCC	0		2.0	V	
	BLU	VLED	-		26.1	V	
		ILED	168			mA	
	TP	VDD					

Note:

1. These range above is maximum value not the actual operating temperature . Actual Operating temperature is no more than 45℃ and temperature refers to the LCM surface temperature ; Length of operation: No more than 8 hours per day, and no more than 4 hours of continuous use one time.
2. BOE is not responsible for product problems beyond the use conditions.

3.0 ELECTRICAL SPECIFICATIONS

3.1 TFT LCD Module

< Table 4. LCD Module Electrical specifications >

Item		Symbol	Unit	Min.	Typ.	Max.	Remark
Input power supply		VSP	V	5.7	5.8	5.9	
		VSN	V	-5.9	-5.8	-5.7	
		VDDD	V	-	-	-	
		IOVCC	V	1.7	1.8	1.9	
Current Spec		I_{Iovcc}	mA	-	TBD	-	
		Sleep I_{Iovcc}	mA	-	-	-	
Frame		Hz		-	15/30/60	-	-
Input Signal Voltage	H Level	V_{IH}	mV	880	-	-	LP MODE
	L Level	V_{IL}	mV	0	-	550	
Output signal Voltage	H Level	V_{OH}	V	1.1	1.2	1.3	
	L Level	V_{OL}	mV	-50	-	50	

Note 1: The operation is guaranteed under the recommended operating conditions only. The operation is not guaranteed if a quick voltage change occurs during operation. To prevent noise, a bypass capacitor must be inserted into the line close to power pin. Please make sure all the design settings are used within this range before the panel is initialed. If IOVCC is not typical value, The Panel Character will not be the best.

Note 2. Please make sure that DC is not supplied to LCD for long period. And do not supply voltage to LCD while within "sleep mode".

3.2 Back-Light Unit

Table 5. LED Driver Electrical Specifications > [Ta =25±2 ℃]

Parameter	Symbol	Values			Unit	Notes
		Min.	Typ.	Max.		
LED Supply Voltage	VLED	-	-	26.1	V	Note 1
	VRP	-	-	-	mV	Ripple
LED Forward Current	ILED	168	-		mA	
Power Consumption	PLED	-	-	4.385	W	
LED Quantity	QLED	-	72	-	EA	
LED Life Time	TLED	10000	-	-	Hrs	Note 2

Notes: 1. PLED = VLED ×ILED (Without LED converter transfer efficiency)

2. The life time of LED, 10,000Hrs, is determined as the time at which luminance of the LED is 50% compared to that of initial value at the typical LED current on condition of continuous operating at 25 ± 2°C.

3.3 INPUT TERMINAL PIN ASSIGNMENT

This LCD employs three interface connections, a 51 pin ZIF connector is used for the LCD module electronics interface, a 11 pin ZIF connector is used for the external backlight system and a 15 pin ZIF connector is used for the internal backlight system.

3.3.1 Pin assignment for LCD module

Connector : 20982-051E-01 (IPEX)

< Table7. Pin Assignment for LCD Module Connector > LCD CN1

Pin No.	Symbol	Description	I/O
1	GND	Ground	P
2	D3-	MIPI differential data3 input (Negative)	I
3	D3+	MIPI differential data3 input (Positive)	I
4	GND	Ground	P
5	D0-	MIPI differential data0 input (Negative)	I
6	D0+	MIPI differential data0 input (Positive)	I
7	GND	Ground	P
8	CLK-	MIPI differential clock input (Negative)	I
9	CLK+	MIPI differential clock input (Positive)	I
10	GND	Ground	P
11	D1-	MIPI differential data1 input (Negative)	I
12	D1+	MIPI differential data1 input (Positive)	I
13	GND	Ground	P
14	D2-	MIPI differential data2 input (Negative)	I
15	D2+	MIPI differential data2 input (Positive)	I
16	GND	Ground	P
17	ID0	System Hardware ID0 Select(Pull Low to VDDI with 10KΩ)	O
18	ID1	System Hardware ID1 Select(Pull Low to GND with 10KΩ)	O
19	LEDPWM_OUT	Output pin for PWM signal of LED driving.	O
20	TE	TE	I
21	REST	This signal will reset the device and must be applied properly to initial ize the chip.	I
22	VDD18	Power Supply 1.8V	P
23	VDD18	Power Supply 1.8V	P
24	NC	NC	-
25	AVDD	Negative input analog power for driver IC	P
26	AVDD	Negative input analog power for driver IC	P
27	NC	NC	-
28	AVEE	Positive input analog power for driver IC	P
29	AVEE	Positive input analog power for driver IC	P
30	TP_INT	Interrupt signal for TP	I
31	TP_RST	The external reset input for TP	I
32	TP_SPI_CS	TP:SPI CS(TYP. 1.8V)-for TP	I
33	TP_SPI_SCLK	TP:SPI SCLK(TYP. 1.8V)-for TP	I
34	TP_SPI_MISO	TP:SPI MISO(TYP. 1.8V)-for TP	O
35	TP_SPI_MOSI	TP:SPI MOSI(TYP. 1.8V)-for TP	I
36	GND	Ground	P
37	D3-	MIPI differential data3 input (Negative)	I
38	D3+	MIPI differential data3 input (Positive)	I
39	GND	Ground	P

3.3.2 Pin assignment for LCD module

Connector : 20982-051E-01 (IPEX)

< Table7. Pin Assignment for LCD Module Connector > LCD CN1

Pin No.	Symbol	Description	I/O
40	D0-	MIPI differential data0 input (Negative)	I
41	D0+	MIPI differential data0 input (Positive)	I
42	GND	Ground	P
43	CLK-	MIPI differential clock input (Negative)	I
44	CLK+	MIPI differential clock input (Positive)	I
45	GND	Ground	P
46	D1-	MIPI differential data1 input (Negative)	I
47	D1+	MIPI differential data1 input (Positive)	I
48	GND	Ground	P
49	D2-	MIPI differential data2 input (Negative)	I
50	D2+	MIPI differential data2 input (Positive)	I
51	GND	Ground	P

3.4 Pin assignment for LED Bar

3.4.1 External BLU System Connector :20655-011E-01

< Table8. Pin assignment for LED Bar >

Pin No	Symbol	Description	I/O
1	FB1	Power for LED1 Cathode	P
2	FB2	Power for LED2 Cathode	P
3	FB3	Power for LED3 Cathode	P
4	FB4	Power for LED4 Cathode	P
5	FB5	Power for LED5 Cathode	P
6	FB6	Power for LED6 Cathode	P
7	FB7	Power for LED7 Cathode	P
8	FB8	Power for LED8 Cathode	P
9	NC	NC	-
10	VLEDB	Power for LED Anode	P
11	VLEDA	Power for LED Anode	P

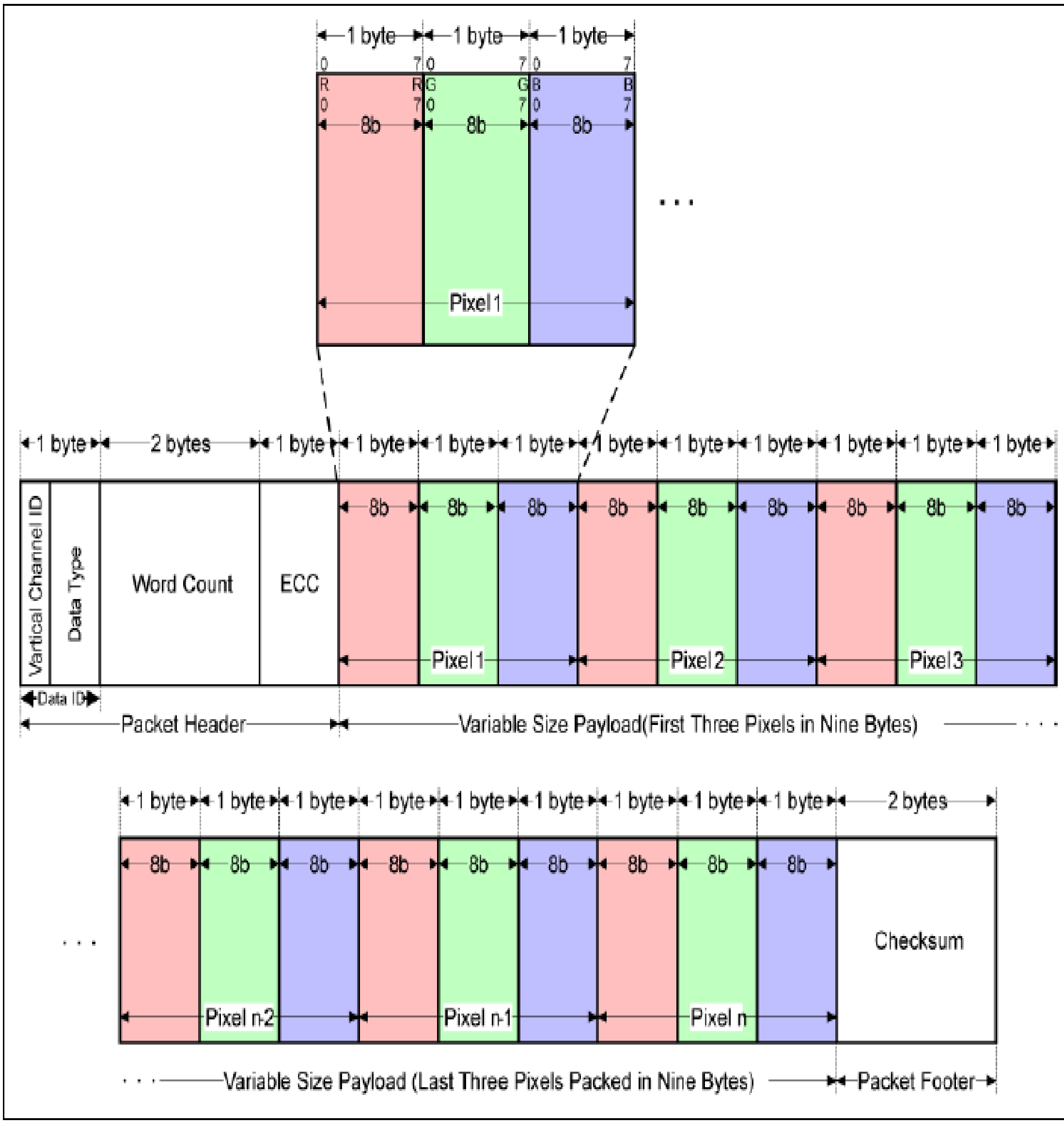
3.4.2 Internal BLU System Connector :20599-015E-01

< Table8. Pin assignment for LED Bar >

Pin No	Symbol	Description	I/O
1	VLEDA	Power for LED Anode	P
2	VLEDA	Power for LED Anode	P
3	VLEDB	Power for LED Anode	P
4	VLEDB	Power for LED Anode	P
5	NC	NC	-
6	NC	NC	-
7	FB1	Power for LED1 Cathode	P
8	FB2	Power for LED2 Cathode	P
9	FB3	Power for LED3 Cathode	P
10	FB4	Power for LED4 Cathode	P
11	FB5	Power for LED5 Cathode	P
12	FB6	Power for LED6 Cathode	P
13	FB7	Power for LED7 Cathode	P
14	FB8	Power for LED8 Cathode	P
15	NC	NC	-

3.4 MIPI Interface Characteristic

3.4.1 Data Format

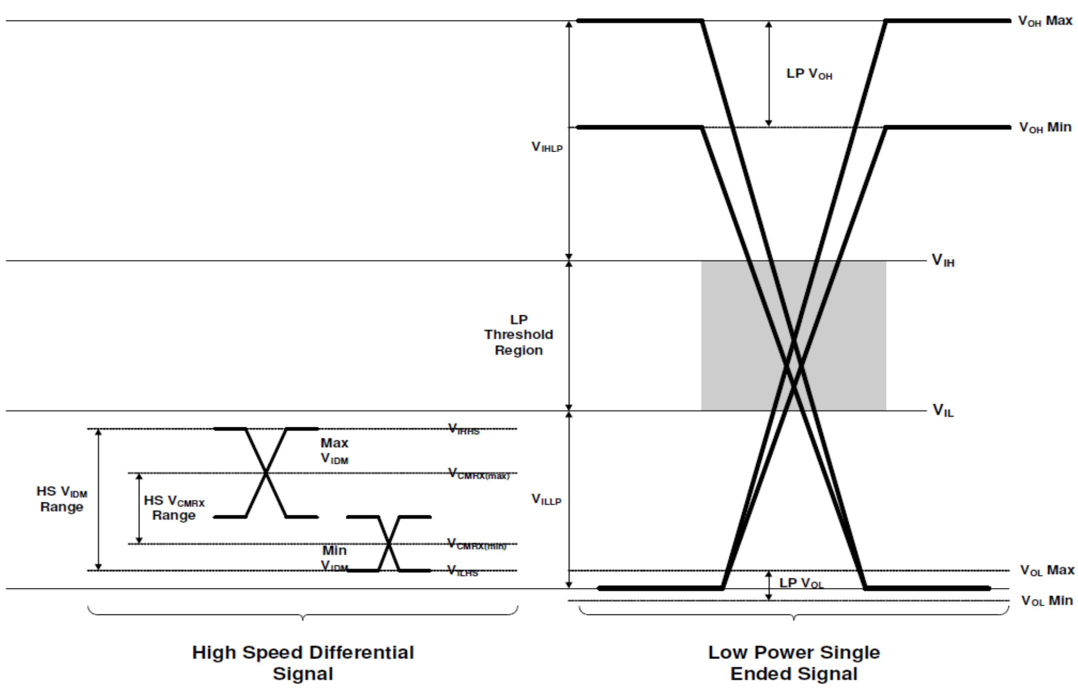


< MIPI Tx Data Configuration >

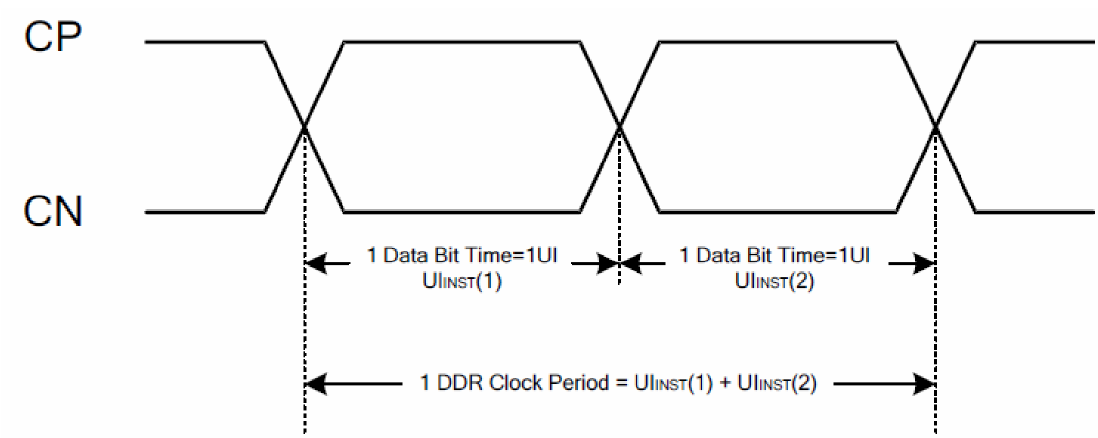
3.4.2 DC Specification

< Table11. DC Specification >

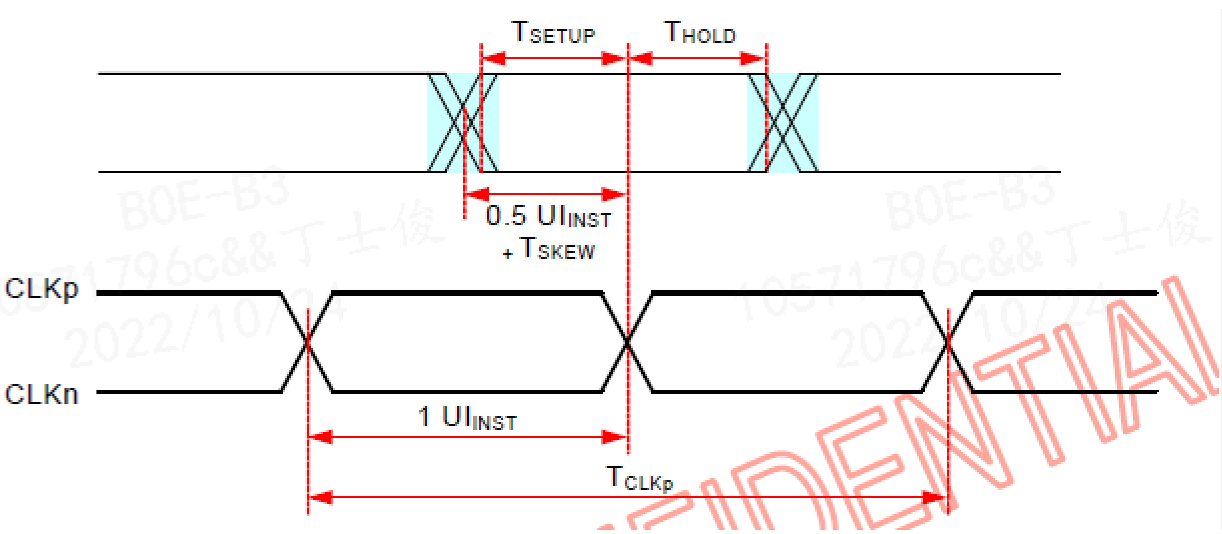
Parameter	**Symbol**	**Min**	**Typ**	**Max**	**Unit**	**Condition**
MIPI digital operation current	I_{VCCIF}	-		-	mA	
MIPI digital stand-by current	$I_{VCCIFST}$	-		-	uA	
MIPI Characteristics for High Speed Receiver						
Single-ended input low voltage	V_{ILHS}	-40	-	-	mV	
Single-ended input high voltage	V_{IHHS}	-	-	460	mV	
Common-mode voltage	V_{CMRXDC}	70	-	330	mV	
Differential input impedance	Z_{ID}	80	100	125	Ω	
HS transmit differential voltage(V_{OD}=V_{DP}-V_{DN})	$\|V_{OD}\|$	140	200	250	mV	
MIPI Characteristics for Low Power Receiver						
Pad signal voltage range	V_I	-50	-	1350	mV	
Ground shift	V_{GNDSH}	-50	-	50	mV	
Output low level	V_{OL}	-50	-	50	mV	
Output high level	V_{OH}	1.1	1.2	1.3	V	



3.4.3 AC Specification

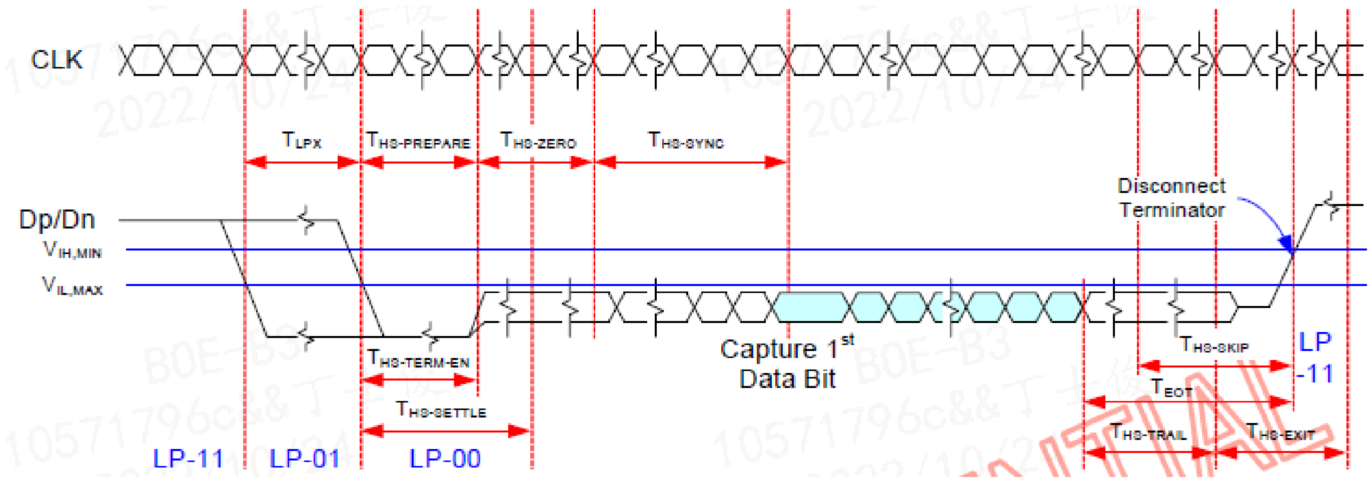


Parameter	Symbol	Min	Typ	Max	Units
UI instantaneous	UI_{INST}	0.833		4	ns



Data to Clock Setup Time [measured at receiver]	T_{SETUP}[RX]	0.15		0.15	UI_{INST}	5
		0.2		0.2	UI_{INST}	6
Data to Clock Hold Time [measured at reciever]	T_{HOLD}[RX]	0.15		0.15	UI_{INST}	5
		0.2		0.2	UI_{INST}	6

3.5 AC Specification

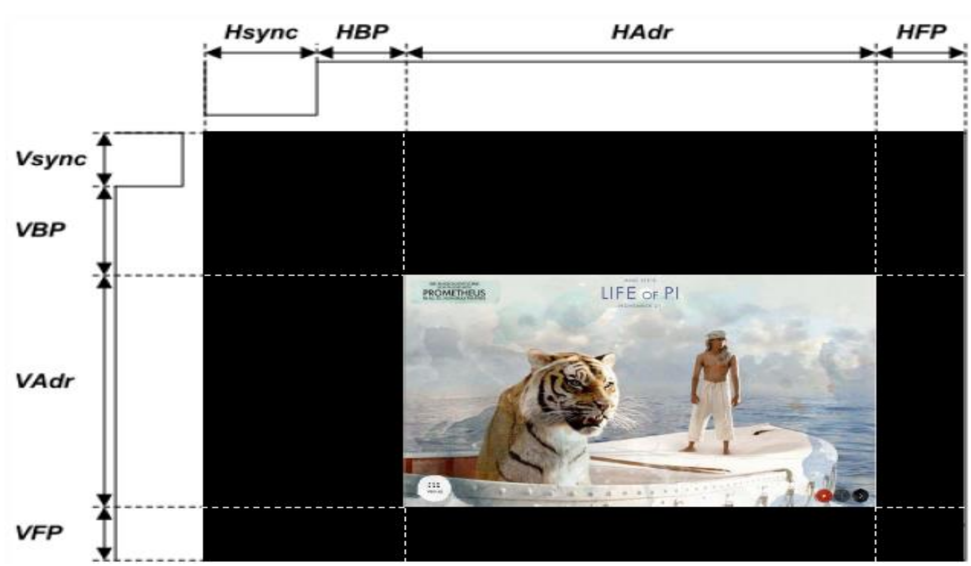


Parameter	Symbol	Min	Typ	Max	Units
Time to drive LP-00 to prepare for HS transmission	$T_{HS-PREPARE}$	40+4UI		85+6UI	ns
Time from start of tHS-TRAIL or tCLK-TRAIL period to start of LP-11 state	T_{EOT}			105+12UI	ns
Time to enable Data Lane receiver line termination measured from when Dn cross VIL,MAX	$T_{HS-TERM-EN}$			35+4UI	ns
Time to drive flipped differential state after last payload data bit of a HS transmission burst	$T_{HS-TRAIL}$	60+4UI			ns
Time-out at RX to ignore transition period of EoT	$T_{HS-SKIP}$	40		55+4UI	ns
Time to drive LP-11 after HS burst	$T_{HS-EXIT}$	100			ns
Length of any Low-Power state period	T_{LPX}	50			ns
Sync sequence period	$T_{HS-SYNC}$		8UI		ns
Minimum lead HS-0 drive period before the Sync sequence	$T_{HS-ZERO}$	105+6UI			ns

3.5 Interface timing Parameter

< Table13. Timing Parameter >

Item			Symbol	min	typ	max	UNIT
LCD	Frame Rate		-		60		Hz
	Pixels Rate		-		373.41		MHz
Timing	DCLK	Frequency	fCLK		-		MHz
		Period	Tclk		-		ns
	Horizontal	Horizontal total time	tHP		3172		t_{CLK}
		Horizontal Active time	tHadr		2944		t_{CLK}
		Horizontal Pulse Width	tHsync		10		t_{CLK}
		Horizontal Back Porch	tHBP		108		t_{CLK}
		Horizontal Front Porch	tHFP		110		t_{CLK}
	Vertical	Vertical total time	tvp		1962		t_H
		Vertical Active time	tVadr		1840		t_H
		Vertical Pulse Width	tVsync		2		t_H
		Vertical Back Porch	tVBP		94		t_H
		Vertical Front Porch	tVFP		26		t_H
Bit Rate			TX SPD (Mbps)		1120		Mbps
Lane				-	8	-	Lane



3.6 Input Color Data Mapping

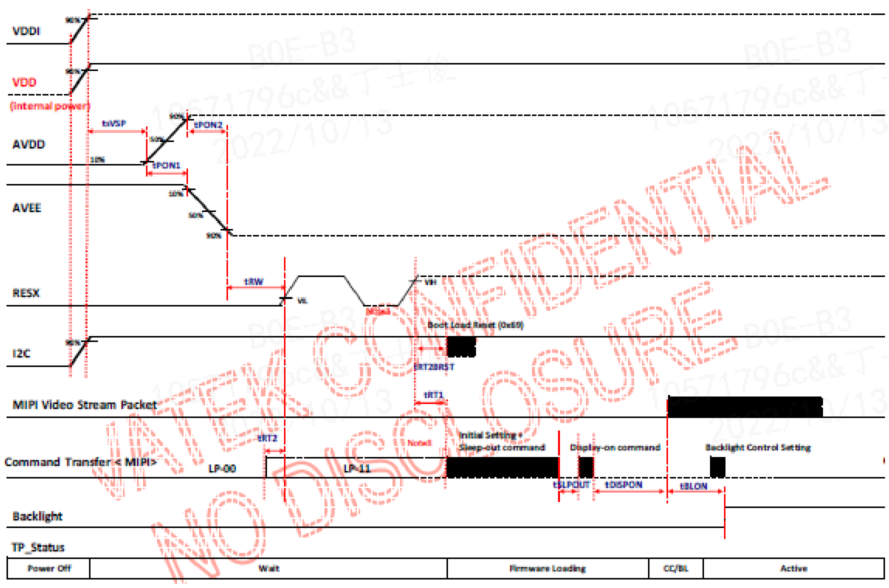
< Table14. Input Signal and Display Color Table >

Color & Gray Scale		Input Data Signal																							
		Red Data								Green Data								Blue Data							
		R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0
Basic Colors	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Blue	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
	Green	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
	Cyan	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Red	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Magenta	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
	Yellow	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
	White	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Gray Scale of Red	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	△	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Darker	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	△				↑								↑								↑				
	▽				↓								↓								↓				
	Brighter	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	▽	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Red	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Gray Scale of Green	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	△	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
	Darker	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
	△				↑								↑								↑				
	▽				↓								↓								↓				
	Brighter	0	0	0	0	0	0	0	0	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0
	▽	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0
	Green	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
Gray Scale of Blue	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	△	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
	Darker	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
	△				↑								↑								↑				
	▽				↓								↓								↓				
	Brighter	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	0	1
	▽	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	0
	Blue	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
Gray Scale of White	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	△	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1
	Darker	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0
	△				↑								↑								↑				
	▽				↓								↓								↓				
	Brighter	1	1	1	1	1	1	0	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	0	1
	▽	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	0
	White	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

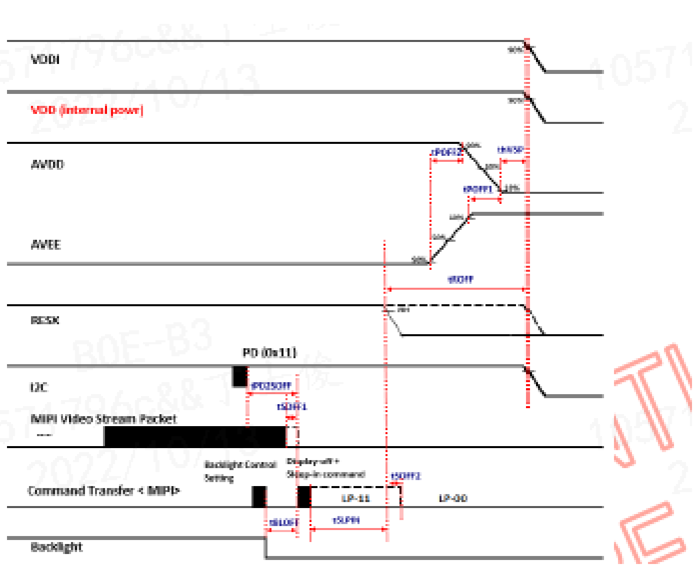
3.7 Power Sequence

To prevent a latch-up or DC operation of the LCD module, the power on/off sequence shall be as shown in below.

Power on sequence



Power off sequence



4.0 OPTICAL SPECIFICATIONS

4.1 Overview

The test of optical specifications shall be measured in a dark room (ambient luminance ≤ 1lux and temperature = 25±2℃) with the equipment of Luminance meter system (CA310 meter system and TOPCON SR-3&RD-80) and test unit shall be located at an approximat e distance 50cm from the LCD surface at a viewing angle of θ and Φ equal to 0°. We refe r to θØ=0 (=θ3) as the 3 o' clock direction (the "right"), θØ=90 (= θ12) as the 12 O ' clock direction ("upward"), θØ=180 (= θ9) as the 9 O' clock direction ("left") and θØ=270(= θ6) as the 6 O' clock direction ("bottom"). While scanning θ and/or Ø, the center of the measuring spot on the Display surface shall stay fixed.

4.2 Optical Specifications

< Table16. Optical Table >

Item	Symbol	Condition	Min	Typ.	Max	Unit	Note
luminance	Bp	θ=0°	320	400	--	cd/m2	center
Brightness Uniformity	△Bp		75	80	--	%	13P
Viewing Angle	θ_L	Cr≥10	80	85	--	deg	Single Center Point
	θ_R		80	85	--		
	ψ_T		80	85	--		
	ψ_B		80	85	--		
Contrast Ratio	Cr	θ=0° FF=0°	1000	1500		-	
Response Time	Tr+Tf		--	--	25	ms	25℃
Color Coordinate of CIE1931	**Rx**	θ=0°	0.642	0.672	0.702	-	穗晶KSF 低蓝光
	Ry		0.291	0.321	0.351		
	Gx		0.232	0.262	0.292		
	Gy		0.648	0.678	0.708		
	Bx		0.105	0.135	0.165		
	By		0.039	0.069	0.099		
	Wx		0.265	0.295	0.325		
	Wy		0.285	0.315	0.345		
NTSC Ratio	NTSC	CIE1931	88	93	--	%	
Color Temperature	CT		-	7778	-		
Flicker	amount	-	-	-	-	dB	
Gamma	-		-	-	-		

Note1:Luminance measurement

The test condition is at ILED=21*2*4mA and measured on the surface of LCD module at 25℃.

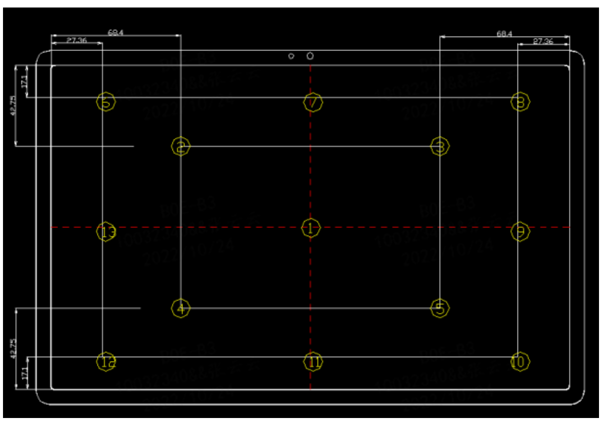
●The data are measured after LEDs are lighted on for more than 5 minutes and LCM displays are fully white. The brightness is the center value of 13 measured spots. Measurement equipment CS2000 or similar equipments(Field of view:1deg,Distance:50cm)

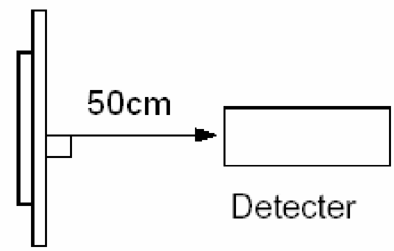
●Measuring surroundings: Dark room.

●Measuring temperature: Ta=25℃.

●Adjust operating voltage to get optimum contrast at the center of the display.

●Measured value at the center point of LCD panel must be after more than 5 minutes while backlight turning on.





Note2:Uniformity

●The test condition is at ILED=21*2*4mA and measured on the surface of LCD module at 25℃.

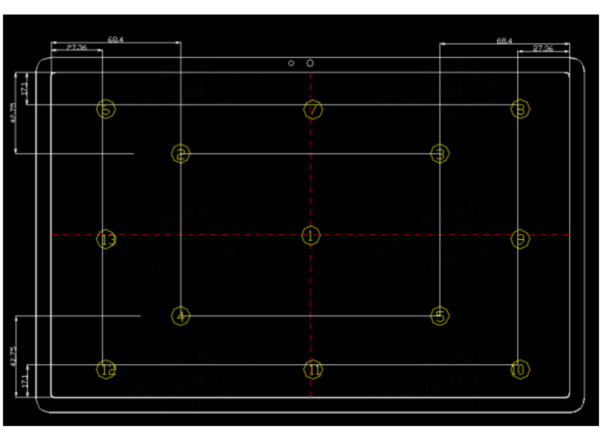
●Measurement equipment:CS2000 or similar equipments

●The luminance uniformity is calculated by using following formula:

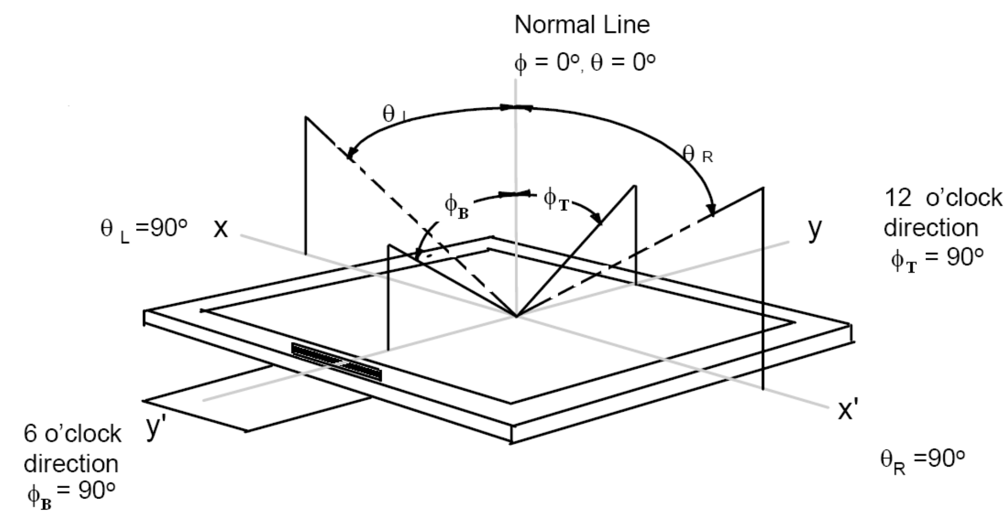
●△Bp = Bp (Min.) / Bp (Max.)×100 (%)

●Bp (Max.) = Maximum brightness in 13 measured spots

●Bp (Min.) = Minimum brightness in 13 measured spots.



Note 3:**The definition of Viewing Angle**
Refer to the graph below marked by θ and Φ.

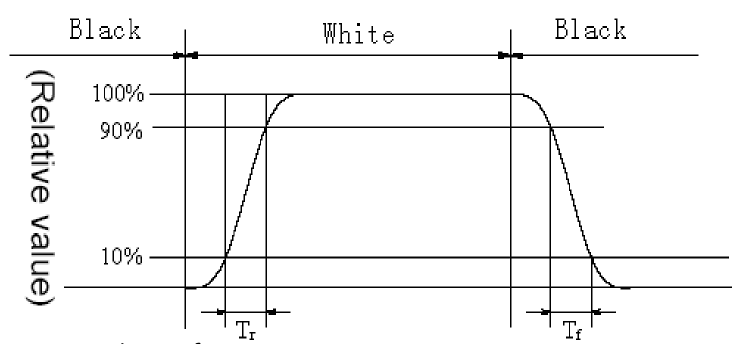


Note4:**ThedefinitionofContrastRatio** (Test LCM using CS2000 or similar equipments):

$$\text{Contrast Ratio(CR)} = \frac{\text{Luminance When LCD is at "White" state}}{\text{Luminance When LCD is at "Black" state}}$$

(Contrast Ratio is measured in optimum common electrode voltage)

Note5:**DefinitionofResponse time.**(Test LCD using DMS501 or similar equipments):
The output sign also photo detector are measured when the input sign also are changed from "black" to "white" (Voltage falling time)and from "white" to "black" (Voltage rising time), respectively . The response time is defined as the time interval between the 10% and 90% of amplitudes . Refer to figures below.



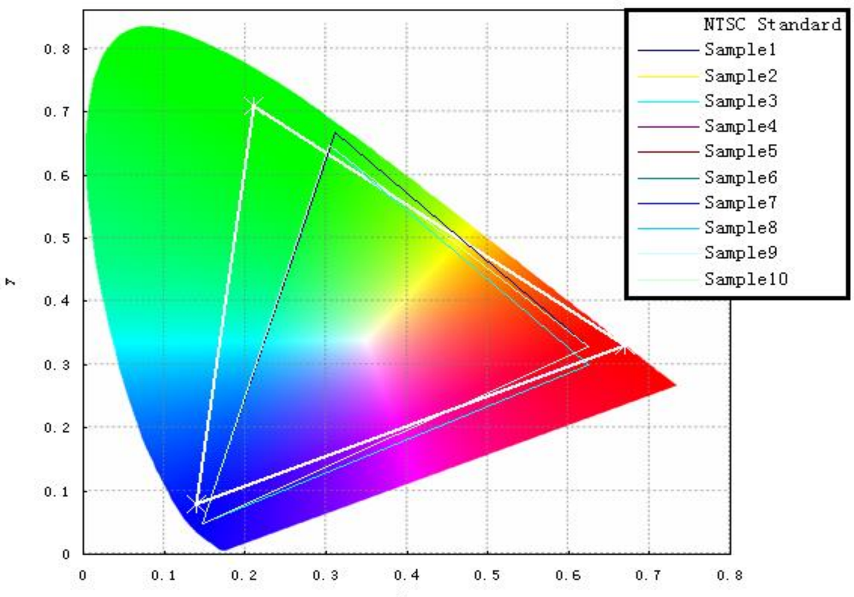
	L0	L1	L2	L3	L4	L5	L6	L7
L0								
L1								
L2								
L3								
L4								
L5								
L6								
L7								

Response time of gray to gray:
Measurement equipment: DMS501 or similar equipments.
Test method: we define 8 grays L0-L7,the grays of L0-L7 were defined as:0,36,73, 109, 146, 182, 219, 255. Theoutputsignalsofphotodetectoraremeasuredwhentheinputsignalsarechanged from "Lx" to "Ly", x, y= [0, 7]. The response time is defined as the time interval between the 10% and 90% of amplitudes. The result of the test can be noted as below:

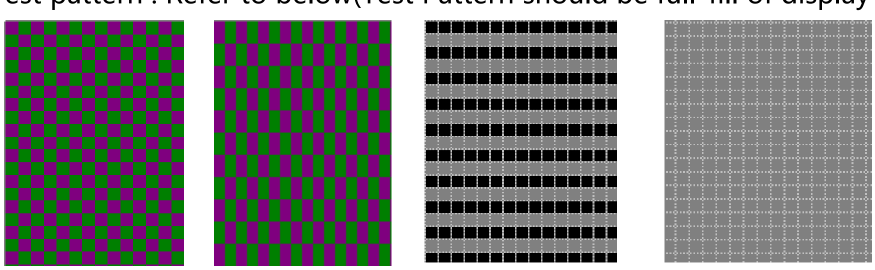
Note 6: Color Coordinates of CIE 1931
The test condition is at ILED=23*5mA and measured on the surface of LCD module at 25℃.
Measurement equipment:CS2000 or similar equipments
The Color Coordinate (CIE 1931) is the measurement of the center of the display shown in below figure.

Note 7: Definition of Color of CIE Coordinate and NTSC Ratio.

$$S = \frac{\text{area of RGB triangle}}{\text{area of NTSC triangle}} \times 100\%$$



Note 8: **Flicker**
●Measurement equipment :CA-310 or similar equipments
●Measuring temperature: Ta=25℃.
●Test method: JEITA method
●Test pattern : Refer to below(Test Pattern should be full-fill of display screen)



1 Dot Inversion, 2 Dot Inversion , Line Inversion , Frame Inversion
The point should be marked is, for line and frame inversion, the background of Flicker Test Pattern
- "gray " are defined as middle gray scale .For example, RGB 24bit "gray" defined as below:

R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0
1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

For Dot inversion, the RGB data for first pixel is (127, 0, 127), the RGB data for the second pixel is (0, 127 , 0).

●Frame Frequency Requirement before test： The LCD must be tuned to more than 65HZ before measu rement.

●Measurement Point: the center of display active area

●Conversion of Flicker ratio:

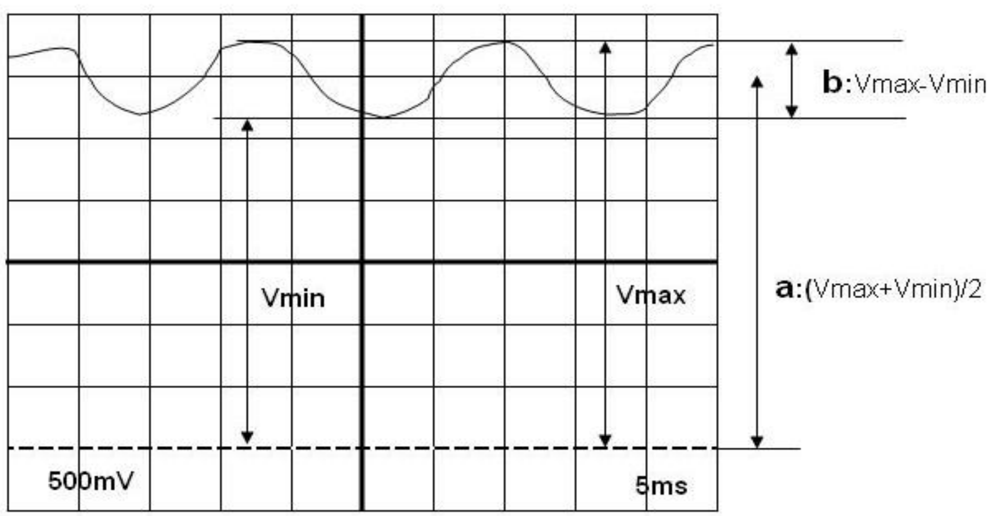
Flicker [dB] = 10 x log[Px/P0]

Where

Px: Maximum power spectrum of AC component after passing through integrator

P0: Power spectrum of DC component after passing through integrator

AC component=b (Refer to below diagram)



Note 9: gamma curve control

●For gamma curve control, Lenovo’s request as below:

●1,the whole curve’s tolerance must control within +/-0.3, Lenovo Iwill test the gray scale below: 0, 8, 16, 25, 33, 41, 49, 58, 66, 74, 82, 90, 99, 107, 115, 123, 132, 140, 148, 156, 165, 173, 181, 189, 19 7,206, 214, 222, 230, 239, 247, 255

Note 10:Crosstalk

●There should be no visible cross-talk in normal direction of the display when the two ” Cross-talk Test Patterns ” below are loaded.

●Measurement equipment:CS2000 or similar equipments

●The point should be marked is, the background of Cross-talk Test Pattern- “gray ” are defined as middle gray scale . For example, RGB 24bit “gray” defined as below:

R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0
1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

5.0 RELIABLITY TEST

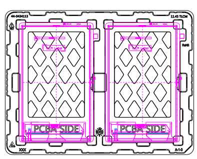
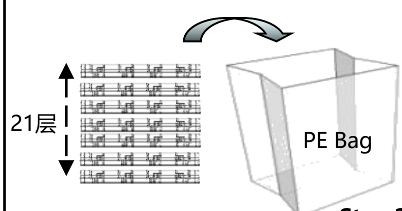
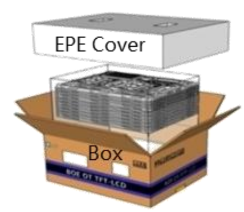
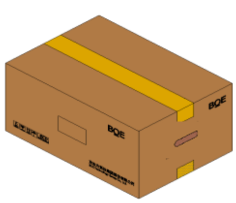
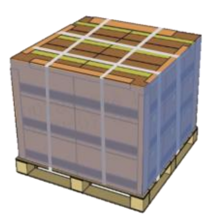
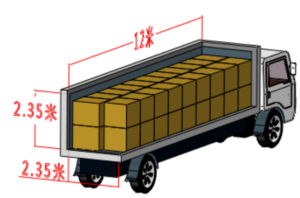
The Reliability test items and its conditions are shown in below.

<Table 17. Reliability Test Parameters >

No	Test Items	Conditions
1	高温驱动（HTO）	60℃，240hr
2	低温驱动（LTO）	-20℃，240hr
3	高温高湿驱动（HTBO）	60℃，90%，240hr
4	高温存储（HTS）	80℃，240hr
5	低温存储（LTS）	-40℃，240hr
6	高温高湿存储（HTBS）	60℃，90%，240hr

6.0 PACKING INFORMATION(产品形态：TLCM)

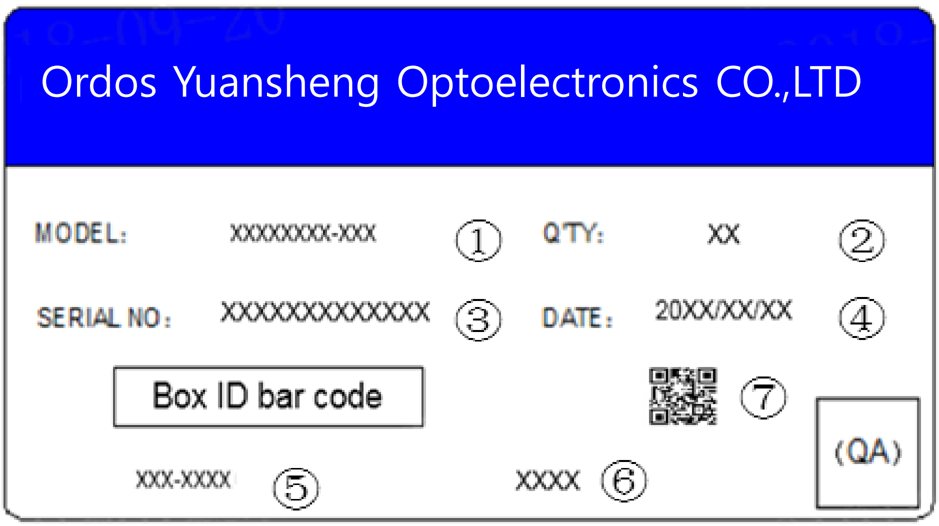
Packing procedure:

在Tray每穴中各放入1pcs TLCM（显示面朝下），再在TLCM上各放置1pcs EPE Spacer 容量：2pcs TCLM/Tray 2pcs EPE Spacer/Tray	堆叠放置20pcs Tray（不旋转堆叠），顶部再放置1pcs 空Tray 将堆叠好的21pcs Tray平放入PE Bag	在Box内放置1pcs EPE Cover 将堆叠好的21pcs Tray平放入Box 再放置1pcs EPE Cover 容量：40pcs TLCM/Box
 **Step 1**	 **Step 2**	 **Step 3**
对Box进行封箱，并粘贴Box标签。	每个Pallet上放3层Box，1层6箱，水平&竖向各放置4根护角，捆绑带“井”字型打包，Pallet外进行缠膜包裹 容量：18pcs Box/Pallet 720pcs TLCM/Pallet	厢车装载方式：一横一竖双层码放 厢车装载量_12m：28800pcs（40托）
 **Step 4**	 **Step 5**	 **Step 6**

6.1 Packing Note

- Box Dimension: 495mm(W) x 395mm(D) x 313mm(H)
- Package Quantity in one Box: 40pcs

6.2 Box label (产品形态：TLCM)

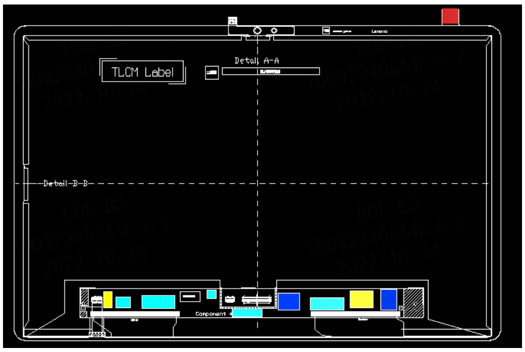


序列号标注部分需打印, 说明如下:

1. FG-CODE(前12位)
2. 产品数量
3. Box ID
4. 包装日期
5. 客户端物料码
6. FG-Code后四位
7. 日期二维码：YYMMDD

Total Size: xxxx mm

7.0 Product Label



Remark:
标签贴附于背板Mark内,
标签尺寸: 48mm × 12mm
1. FG-CODE: TS127QFM-LL0
2. MDL ID
3. 8S码, 客户料号: **SD18D76794**
4. 8S 码对应二维码
5. MDL ID 对应条纹码

MDL ID 编码规则

序列号	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
代码	L	X	A	6	X	X	X	X	K	P	0	X	X	X	X	X	X
描述	GBN代码		等级	B6	年份		月	FG Code后四位				序列号					

8S 编码规则

序列号	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
代码	8	S	S	D	1	8	D	7	6	7	9	4	**A**	6	Y	S	Y	M	D	x	x	x	x
描述	固定值		产品客户端物料号										版本号	代码	供应商产地		年月日			序列号			

8.0 Handling & Cautions

8.1 Mounting Method

- The panel of the LCD consists of two thin glasses with polarizers which easily get damaged. So extreme care should be taken when handling the LCD.
- Excessive stress or pressure on the glass of the LCD should be avoided. Care must be taken to insure that no torsional or compressive forces are applied to the LCD unit when it is mounted.
- If the customer's set presses the main parts of the LCD, the LCD may show the abnormal display. But this phenomenon does not mean the malfunction of the LCD and should be pressed by the way of mutual agreement.

8.2 Caution of LCD Handling and Cleaning

- Since the LCD is made of glass, do not apply strong mechanical impact or static load onto it. Handling with care since shock, vibration,and careless handling may seriously affect the product. If it falls from a high place or receives a strong shock, the glass may be broken.
- The polarizers on the surface of panel are made from organic substances. Be very careful for chemicals not to touch the polarizers or it leads the polarizers to be deteriorated.
- If the use of a chemical is unavoidable, use soft cloth with solvent (recommended below) to clean the LCD 's surface with wipe lightly.
  -IPA(Isopropyl Alcohol), Ethyl Alcohol, Trichlorotriflorothane
- Do not wipe the LCD's surface with dry or hard materials that will damage the polarizers and others. Do not use the following solvent.
  -Water, Ketone, Aromatics
- It is recommended that the LCD be handled with soft gloves during assembly, etc. The polarizers on the LCD's surface are vulnerable to scratch and thus to be damaged by sharp particles.
- Do not drop water or any chemicals onto the LCD's surface.
- A protective film is supplied on the LCD and should be left in place until the LCD is required for operation.
- The ITO pad area needs special careful caution because it could be easily corroded. Do not contact the ITO pad area with HCFC,Soldering flux,Chlorine,Sulfur,saliva or fingerprint. To prevent the ITO corrosion, customers are recommended that the ITO area would be covered by UV or silicon.

8.3 Caution Against Static Charge

- The LCD modules use C-MOS LSI drivers, so customers are recommended that any unused input terminal would be connected to Vdd or Vss, do not input any signals before power is turn on, and ground you body, work/assembly area, assembly equipments to protect against static electricity.
- Remove the protective film slowly, keeping the removing direction approximate 30-degree not vertical from panel surface, If possible, under ESD control device like ion blower, and the humidity of working room should be kept over 50%RH to reduce the risk of static charge.
- Avoid the use work clothing made of synthetic fibers. We recommend cotton clothing or other conductivity-treated fibers.
- In handling the LCD, wear non-charged material gloves. And the conducting wrist to the earth and the conducting shoes to the earth are necessary.

8.4 Caution For operation

- It is indispensable to drive the LCD within the specified voltage limit since the higher Voltage than the limit causes the shorter LCD's life. An electro-chemical reaction due to DC causes undesirable deterioration of the LCD so that the use of DC drive should avoid.
- Do not connect or disconnect the LCD to or from the system when power is on.
- Never use the LCD under abnormal conditions of high temperature and high humidity.
- When expose to drastic fluctuation of temperature (hot to cold or cold to hot) ,the LCD may be affected; Specifically, drastic temperature fluctuation from cold to hot ,produces dew on the LCD's surface which may affect the operation of the polarizer and the LCD.
- Response time will be extremely delayed at lower temperature than the operating temperature range and on the other hand at higher temperature LCD may turn black at temperature above its operational range. However those phenomena do not mean malfunction or out of order with the LCD. The LCD will revert to normal operation once the temperature returns to the recommended temperature range for normal operation.
- Do not display the fixed pattern for a long time because it may develop image sticking due to the LCD structure. If the screen is displayed with fixed pattern, use a screen saver.

8.5 Packaging

- Modules use LCD element, and must be treated as such.
  -Avoid intense shock and falls from a height.
  -To prevent modules from degradation, do not operate or store them exposed directly to sunshine or high temperature/humidity for long periods.

8.6 Storage

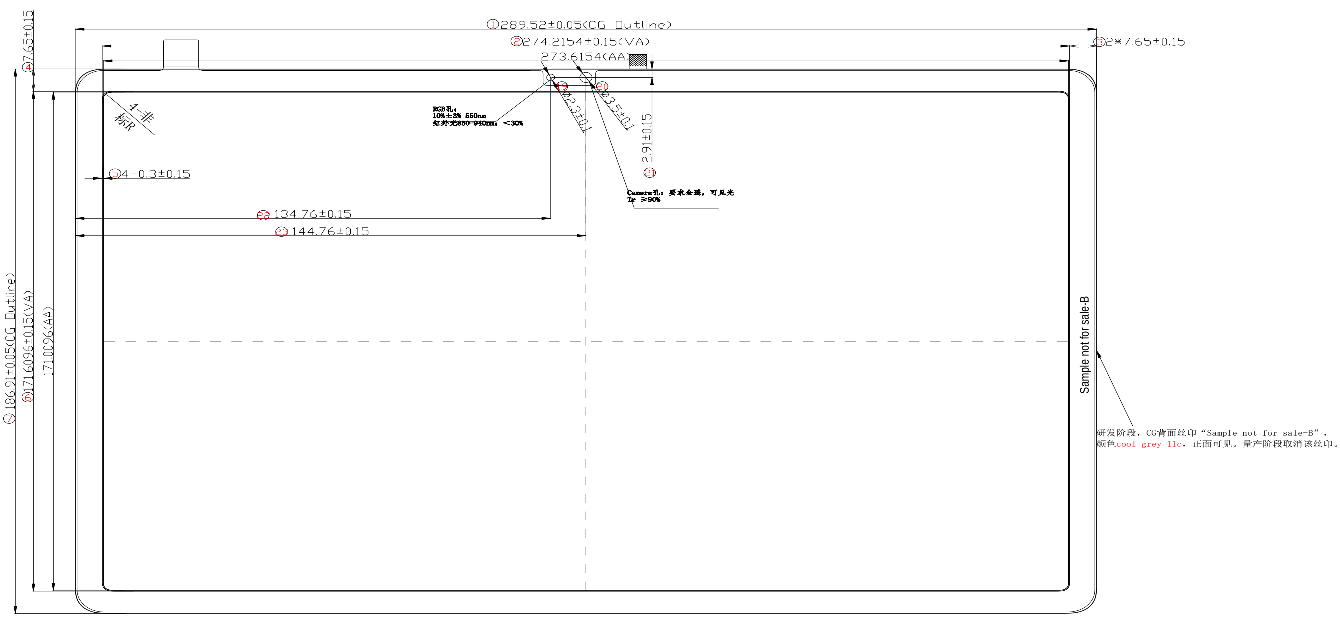
- A slight dew depositing on terminals is a cause for electro-chemical reaction resulting in terminal open circuit. Relative humidity of the environment should therefore be kept below 60%RH.
- Original protective film should be used on LCD' s surface (polarizer). Adhesive type protective film should be avoided, because it may change color and/or properties of the polarizers.
- Do not store the LCD near organic solvents or corrosive gasses.
- Keep the LCD safe from vibration, shock and pressure.
- Black or white air-bubbles may be produced if the LCD is stored for long time in the lower temperature or mechanical shocks are applied onto the LCD.
- In the case of storing for a long period of time for the purpose or replacement use, the following ways are recommended.
  -Store in a polyethylene bag with sealed so as not to enter fresh air outside in it.
  -Store in a dark place where neither exposure to direct sunlight nor light is.
  -Keep temperature in the specified storage temperature range.
  -Store with no touch on polarizer surface by the anything else. If possible, store the LCD in the packaging situation LCD when it was delivered.

8.7 Safety

- For the crash damaged or unnecessary LCD, it is recommended to wash off liquid crystal by either of solvents such as acetone and ethanol an should be burned up later.
- In the case the LCD is broken, watch out whether liquid crystal leaks out or not. If your hands touch the liquid crystal, wash your hands cleanly with water an soap as soon as possible.
- If you should swallow the liquid crystal, first, wash your mouth thoroughly with water, then drink a lot of water and induce vomiting, and then, consult a physician.
- If the liquid crystal should get in your eyes, flush your eyes with running water for at least fifteen minutes.
- If the liquid crystal touches your skin or clothes, remove it and wash the affected part of your skin or clothes with soap and running water.

9.0 APPENDIX

Mechanical Drawing
Drawing Attachment: Front



Mechanical Drawing
Drawing Attachment: Back